

Mini Project Question: Multi-Armed Bandit

Using **Python**, implement a **Multi-Armed Bandit** problem in which an agent must choose between several arms with unknown reward probabilities.

Your project must:

1. Create a **5-arm bandit** where each arm has a different unknown reward probability.
2. Use an **ϵ -greedy strategy** to balance **exploration and exploitation**.
3. Run the experiment for at least **1,000 trials**.
4. Record the reward obtained at each trial and calculate the **total and average reward**.
5. Plot the **cumulative reward** and **average reward** using Python/Matplotlib.
6. Compare different values of ϵ , such as **0.01, 0.1, and 0.5**.
7. Identify which arm is selected most often and explain why.
8. Briefly discuss how **exploration and exploitation** affect the agent's performance.

Submission

Please submit the following:

- Python code
- Reward graphs
- Results for the different ϵ values
- A short analysis and discussion of your results

Deadline: September 30, 2026

Please submit your project by email to: hoda.zamani@khazar.org

Your email should include the following information:

- **Student Name**
- **Student ID**
- **Field of Study / Major**

Thank you!